

PEIYU LIU

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2004 Mowry Road, Gainesville, FL

EDUCATION

University of Florida

Ph.D. in Biostatistics

Dissertation: []

Advisors: Matt Hitchings

Gainesville, U.S.

Aug. 2023 –

Shaanxi Normal University

Master of Science in Computational Mathematics

Thesis: *Multi-Wave Dynamic Modeling of COVID-19 Control: From Non-Pharmaceutical Interventions to Vaccine-Based Normalization*

Advisor: Sanyi Tang

Xian, China

Aug. 2018 – Jun. 2021

Shaanxi Normal University

Bachelor of Science in Statistics

Thesis: *Smart Campus Shuttle Optimization via Agent-Based Modeling: A Simulation Study of Shaanxi Normal University*

Advisor: Sanyi Tang

Xian, China

Aug. 2013 – Jun. 2017

RESEARCH INTERESTS

Methodology: Bayesian hierarchical, Agent-based modeling, Stochastic processes (Poisson processes, Partially Observed Markov Processes, Diffusion processes), Neural networks, Survival analysis, Causal inference **Application:** Clinical trials, Cluster randomized vaccine trials, Molecular epidemiology

PROFESSIONAL EXPERIENCE

Biostatistician

Biostatistics Core, Emerging Pathogens Institute / University of Florida

Gainesville, U.S.

May 2025 – Present

- Led statistical methodology development and biostatistical support across multiple infectious disease and vaccine research projects, delivering analyses via presentations and summary reports to interdisciplinary stakeholders.
- Developed and released the open-source R package *vaxineR* for outbreak risk modeling (R_e estimation, vaccination scenario simulation, county-level visualization), and quantified outbreak probabilities using 10 years of Florida measles vaccination data; findings were featured in *Science News*.
- Collaborated with clinicians and virologists on RSV severity and pediatric clinical-genomic studies, applying Cox proportional hazards and multivariable regression models to EHR and genotype data; contributed figures and insights to a manuscript submitted to *Nature Communications*.

Research Fellow and Lecturer

Department of Mathematics / Hunan Institute of Science and Technology

Changsha, China

Jun 2021 – Aug 2023

- Taught Probability Theory and Mathematical Statistics to 200+ students (243 class hours/year) while leading independent research in dynamical systems modeling of tumor-immune interactions, including bifurcation and sensitivity analyses to guide radioimmunotherapy optimization.
- Applied machine learning methods (EM algorithm, Lasso regression, GAMs) and spatial bivariate choropleth analysis to multinational COVID-19 datasets, assessing weather-transmission associations across 10 countries; directed a ¥50,000 municipal science grant as Project Director.

PUBLICATIONS

Methodology Publications

1. **Liu, P.**, Rong, L., Cummings, D., and Hitchings, M. (2026). "Mosquito salivary protein antibodies as efficient endpoints for vector control trials: A mechanistic comparison with clinical and entomological outcomes," *Infectious Disease Modelling*, In Press.
2. **Liu, P.**, He, S., Rong, L., and Tang, S. (2020). "The effect of control measures on COVID-19 transmission in Italy: Comparison with Guangdong province in China," *Infectious Diseases of Poverty*, 9(130): 1–13.
3. Tang, B., **Liu, P.**, Yang, J., Wu, J., Tang, S., and Xiao, Y. (2020). "The challenges of the coming mass vaccination and exit strategy in prevention and control of COVID-19, a modelling study," *medRxiv*.
<https://doi.org/10.1101/2020.12.18.20248478>

Application Publications

1. Yan, Q., and **Liu, P.** (2023). "Effects of climatic factors and control measures on the spread of COVID-19," *Journal of Zhejiang University (Science Edition)*, 50(2).
2. Xia, F., Xiao, Y., **Liu, P.**, and Cheke, R. (2020). "Differences in how interventions coupled with effective reproduction numbers account for marked variations in COVID-19 epidemic outcomes," *Mathematical Biosciences and Engineering*, 17: 85–98.
3. Wang, G., Ding, W., Zhang, Z., and **Liu, P.** (2022). "Qualitative analysis of a tumor immune model with impulsive control strategies at fixed time," *Journal of Hunan Institute of Science and Technology*, 35(3): 1–6.

Submitted Papers

1. Santos, L. T., **Liu, P.**, [et al.]. "Increased genomic diversification frequency of Human Respiratory Syncytial Virus (HRSV) type A in a pediatric cohort in the United States after the onset of the COVID-19 pandemic," Submitted to *Microorganisms*, under review.

In Preparation

1. Robledo, J., Izquierdo, G., [et al.], and **Liu, P.** "A multi-site field study demonstrates that integrated biosecurity delays Fusarium wilt of banana spread in multi-host landscapes." In preparation.
2. Robledo, J., **Liu, P.**, Vallad, G., and Garrett, K. "Filling the Gaps Between Host Availability Raster Maps with Expert Elicitation to Generate Uncertainty Maps for Epidemiological Studies: Bacterial Spot of Tomato in Florida as a Case Study." In preparation.

PRESENTATIONS

University of Florida Public Health and Health Professions Research Days (PHHP Days 2026)	Gainesville, FL
A non-homogeneous Poisson process model linking serological biomarkers of Aedes aegypti bite exposure to vector control intervention efficacy (oral presentation).	April 2026
University of Florida Emerging Pathogens Institute Research Days (EPI Days 2026)	Gainesville, FL
Mosquito Bite Exposure Biomarkers as Effective Endpoints for Cluster randomized trials (poster presentation).	April 2026
2026 Annual ASA Florida Chapter Meeting	Miami, FL
A Unified Computational Framework for Evaluating Vector Control Trial Endpoints:	March 2026

Nonhomogeneous Poisson Modeling and an Interactive R Shiny Implementation (poster presentation).

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID 2025)

Atlanta, GA

A Unified Computational Framework for Evaluating Vector Control Trial Endpoints:
Nonhomogeneous Poisson Modeling and an Interactive R Shiny Implementation (oral presentation).

July 2025

University of Florida, Department of Biostatistics Student Seminar

Gainesville, FL

A Statistical Learning Framework for Exploratory Analysis and Time Series Modeling of U.S. Weather Data.

April 2024

TEACHING

PHC 6059 Applied Survival Analysis, Teaching Assistant

Fall 2025

Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID), Instructor

Summer 2025

MAA 350 Probability Theory and Mathematical Statistics, Instructor

Fall 2021 – Spring
2023

STAT 538 Multivariable Regression, Instructor

Fall 2022

STAT 539 Statistics Principles, Instructor

Spring 2022

HONORS AND AWARDS

PhD Student Travel Award, Summer Institute in Statistics and Modeling in Infectious Diseases, 2025.

Research and teaching fellowship, University of Florida, 2023–

Meritorious Mentor Award (for advising a Second Prize-winning team), Mathematical Modeling Competition, Chinese Society for Industrial and Applied Mathematics (CSIAM), 2022.

National Scholarship, Ministry of Education of China (MOE), 2021 (top 0.5% nationwide).

Second Prize (as student competitor), Mathematical Contest in Modeling (MCM), Chinese Society for Industrial and Applied Mathematics (CSIAM), 2015.

First/Second Scholarship, Department of Mathematics and Statistics, Shaanxi Normal University, 2013–2017.

APPLICATIONS/PACKAGES

R Package – vxineR: Statistical toolkit for vaccine-coverage analysis and outbreak-risk modeling. Available on [CRAN](#).

R-Shiny Apps:

i. Oncology Dose Optimization (Clinical Utility Score framework, exposure-response modeling, survival analysis): <https://peiyuliu.shinyapps.io/ontology-cus-pkboin-v1/>

ii. Dengue Clinical Trial (serological vs. clinical endpoints): <https://peiyuliu.shinyapps.io/anpp-trial-comparison/>

iii. US Weather Analyses (ggplot2 visualization, K-means clustering, Machine Learning forecasting): <https://peiyuliu.shinyapps.io/us-weather-analysis/>

iv. SIR Model Estimation (Least Squares, Bootstrap, MCMC): <https://peiyuliu.shinyapps.io/sir-estimation-comparison/>

ACADEMIC SERVICES
